Lu Zhang | VFX 721 | Spring 2026

Project 02 | Complex Scene

Flash in the Fronds - User Guide and Breakdown

Render Statistics

Renderer	Karma (BRAY_HdKarma)
Houdini Version	21.0.440
Average Render Time	1h 39m 44s
Image Resolution	1080 × 1620

Geometry Statistics

Polygons (total)	39,104,498
Curves (total)	1,198,800
Volumes	1

Lights

Total Lights	4
Dome (HDRI)	1
Distant	1
Point	1
Rectangle	1

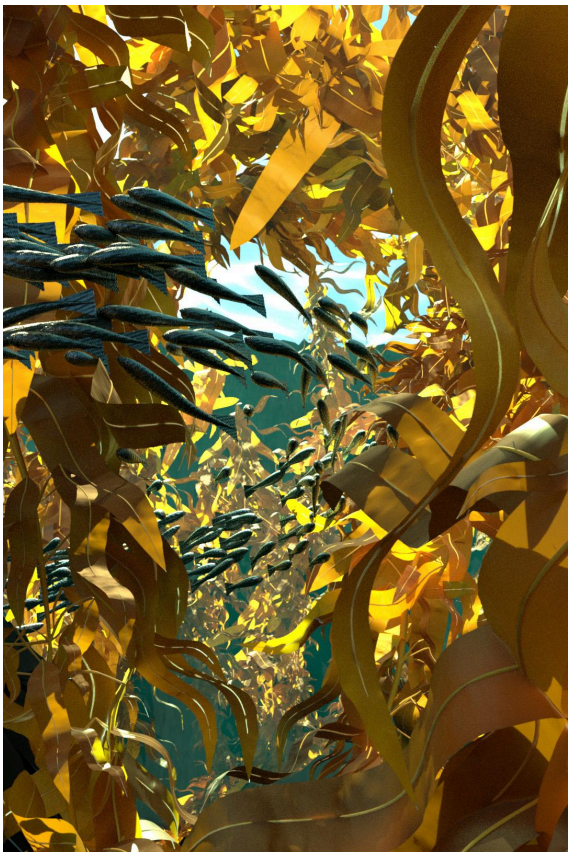
Sampling

Max Samples / Pixel	256
Noise Value	0.01
Min Rays / Pixel	−1 (adaptive)
Variance AA Min / Max	1 / 9
Geometry Samples	2

Project Description

An underwater kelp forest corridor filled with dense golden kelp leaves framing the scene from above and both sides, while a tightly packed school of sardines flows through the center in layered formations. Warm sunlight filters through the canopy, creating strong volumetric light beams, shimmering caustics, and deep atmospheric falloff toward the blue water in the background, with smaller fish drifting between the hanging kelp strands.



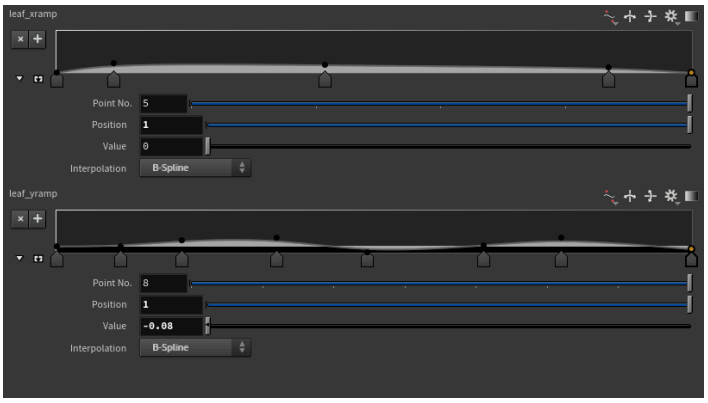


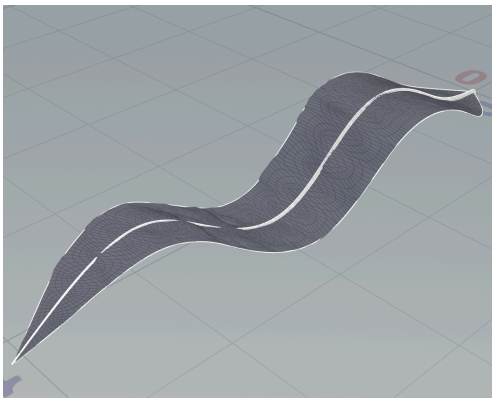
Left: reference: https://pin.it/3FykAj8vr *Right: render*

Process

1. **Kelp:** The kelp leaf was created procedurally using two ramp controls to shape the geometry. The first ramp controlled the overall width and taper of the leaf, helping form the main silhouette from base to tip. The second ramp added subtle waviness and irregular deformation to make the leaf feel more natural and organic. By adjusting

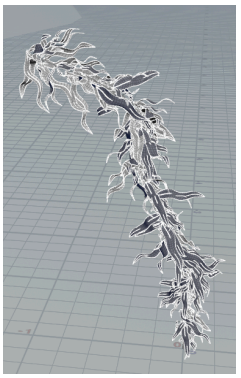
the ramp values and interpolation, I could quickly create different kelp variations while keeping the setup fully procedural.





Then I imported the procedural leaf into an L-system setup to generate the kelp structure.



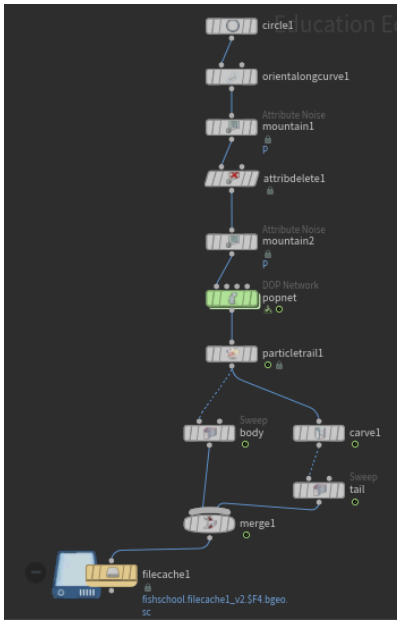


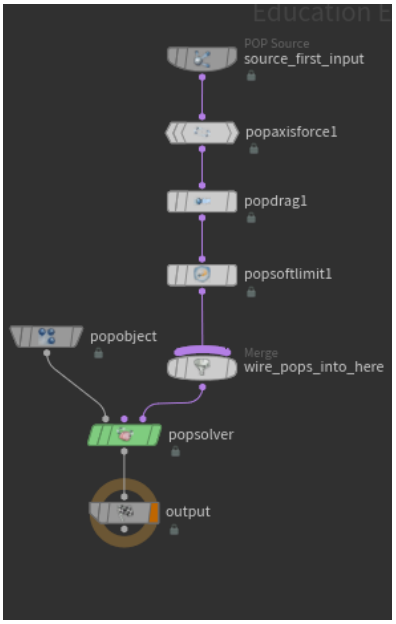
After scattering points across the height field, I assigned a random variant (i@variant = int(rand(@ptnum) * 5)) attribute to each point using a Point Wrangle. This attribute was used in the Copy to Points node to randomly distribute different kelp variations across the environment.

2. **Fish school**

The sardine school was created using a particle-based setup driven by POP simulation. I started by generating particles from a circular source shape, then used POP forces such as drag and axis force to control the swirling motion and group

behavior. Attribute Noise nodes were added before the simulation to introduce more



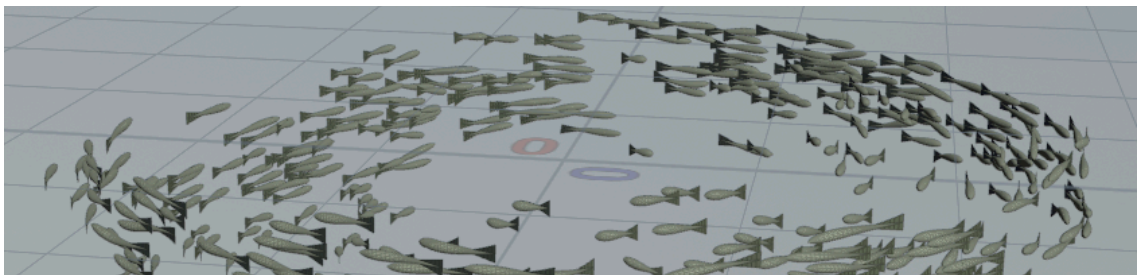


natural variation in movement and spacing.

After simulating the particles, I generated trails using a particle trail node and used sweep operations to build the fish body and tail geometry procedurally along the curves. This allowed the fish to inherit smooth directional movement directly from the simulation while maintaining variation across the school.

3. Environment

For the environment, I used a HeightField to create the underwater terrain. I also used a grid with a Mountain node, then extruded it to create the water volume. To enhance the sense of depth, I added VDB fog, which helped create stronger atmospheric falloff and made the background feel more distant.



BIBLIOGRAPHY AND REFERENCES

- https://pin.it/3FykAj8vr